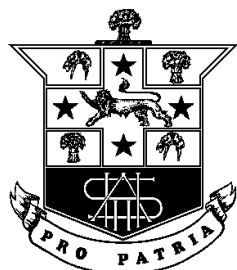


STUDENT'S NAME: _____

TEACHER'S NAME: _____



2019

TRIAL HIGHER
SCHOOL
CERTIFICATE
EXAMINATION

Mathematics Extension 1

Assessment Task 4

Examiners

- Mrs Biczó
- Ms Crancher
- Mr Potaczala
- Ms Tarannum

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- NESA-approved calculators may be used
- A Reference sheet is provided for your use
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks: 70

Section I – 10 marks (pages 2 – 4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 5 – 10)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section 1**10 marks****Attempt Questions 1 – 10****Allow about 15 minutes for this section.**

Use the multiple choice answer sheet provided for Questions 1 – 10

1. Which of the following is an expression for $\int 2 \sin^2 x \, dx$?

A. $x + \frac{1}{2} \sin 2x + c$

B. $x - \frac{1}{2} \sin 2x + c$

C. $x + \sin 2x + c$

D. $x - \sin 2x + c$

2. The general solution for $\cos \theta = -\frac{1}{2}$ is given by which of the following?

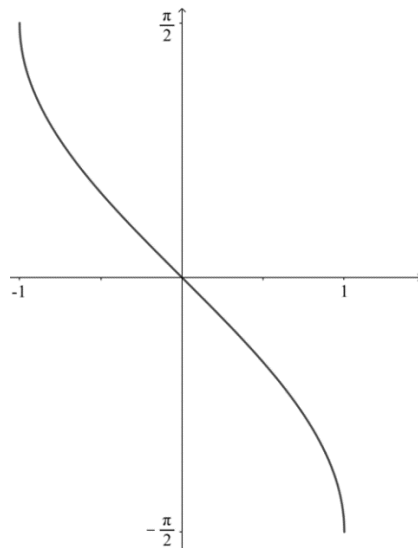
A. $\theta = 2n\pi \pm \frac{\pi}{3}$, where n is real

B. $\theta = 2n\pi \pm \frac{\pi}{3}$, where n is an integer

C. $\theta = 2n\pi \pm \frac{2\pi}{3}$, where n is real

D. $\theta = 2n\pi \pm \frac{2\pi}{3}$, where n is an integer

3. Which of the following functions best describes the graph below?



A. $y = \sin^{-1} x$

B. $y = -\sin^{-1} x$

C. $y = \cos^{-1} x$

D. $y = -\cos^{-1} x$

8. When the polynomial $P(x) = x^3 + ax^2 + 7$ is divided by $x + 2$, the remainder is 11. What is the value of a ?
- A. 3 B. -3 C. -4 D. 26
9. What is the solution to the inequality $\frac{x^2 - 4}{2x} < 0$?
- A. $x < -2$ and $0 < x < 2$ B. $-4 < x < 0$ and $x > 2$
- C. $-2 < x < 0$ and $x > 2$ D. $-2 < x < 0$ and $x > 4$
10. In how many ways can 5 men and 5 women be arranged around a circular table if the women and men are to alternate?
- A. 600 B. 2 880 C. 14 400 D. 86 400

End of Section 1

Section 2**60 marks****Attempt Questions 11 – 14****Allow about 1 hours and 45 minutes for this section****Answer each question in a separate answer booklet.**

In questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new answer booklet.**Marks**

- (a) Prove that $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \frac{\theta}{2}$ **2**
- (b) (i) Write $\cos x - \sqrt{3} \sin x$ in the form $r \cos(x + \alpha)$ **2**
- (ii) Hence, find the maximum value of $\cos x - \sqrt{3} \sin x$ and the first positive value of x for which this occurs. **2**
- (c) From the top of a vertical tower, of height h , the angle of depression to a man M , standing due south of the base O of the tower, is 42° . From the top of the tower, another man G is observed with angle of depression 32° . The men are standing 500 metres apart, with G due east of M . Find the height h of the tower, to the nearest metre. **3**
- (d) The curves $f(x) = \sin x$ and $g(x) = \cos x$ meet at $x = \frac{\pi}{4}$. It is given that **2**
- $f'\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ and $g'\left(\frac{\pi}{4}\right) = \frac{-1}{\sqrt{2}}$. Calculate the acute angle between the curves at $x = \frac{\pi}{4}$. Give your answer in radians to two decimal places.

Question 11 is continued on the next page

Question 11 continued

(e) (i) Show that $\tan(x+h) - \tan x = \frac{\sin h}{\cos(x+h)\cos x}$ **2**

(ii) Hence, using the first principles definition of the derivative, **2**
 $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, show that if $f(x) = \tan x$, then $f'(x) = \sec^2 x$.

Question 12 is on the next page

Question 12 (15 marks) Start a new answer booklet.**Marks**

- (a) Consider the function $y = \tan^{-1} \frac{1}{x}$, where $x \neq 0$:
- (i) Find $\frac{dy}{dx}$ 2
- (ii) Show that $\frac{d}{dx} \left(\tan^{-1} x + \tan^{-1} \frac{1}{x} \right) = 0$ 1
- (iii) Hence, or otherwise, sketch $y = \tan^{-1} x + \tan^{-1} \frac{1}{x}$ for $x > 0$ 1
- (b) Using the substitution $u = \sqrt{x}$, find $\int_1^9 \frac{dx}{x + \sqrt{x}}$. 3
- (c) Show that the polynomial $P(x) = 2x^3 - 5x^2 - 9x + 18$ has a root x such that $1 < x < 2$. 2
- (d) In a town, the annual growth rate of the population N is given by $\frac{dN}{dt} = k(N - 125)$, where k is a constant.
- (i) Show that $N = 125 + Ae^{kt}$ is a solution to $\frac{dN}{dt} = k(N - 125)$ 2
- (ii) If the initial population was 25,650, find k given the population was 31,100 after 5 years. Round your answer to three decimal places. 2
- (e) Sketch the graph $y = \frac{x^2}{4 - x^2}$, clearly showing any asymptotes or intercepts with the co-ordinate axes. Your graph should be at least a quarter of a page. 2

- (a) Prove that for all positive integers n , $9^{n+2} - 4^n$ is divisible by 5. 3
- (b) (i) By using the binomial expansion, show that 2
- $$(a+b)^n - (a-b)^n = 2 \binom{n}{1} a^{n-1} b + 2 \binom{n}{3} a^{n-3} b^3 + 2 \binom{n}{5} a^{n-5} b^5 + \dots$$
- (ii) What is the last term in the expansion when n is even? 1
Give your answer in simplest form.
- (c) A fair six-sided die is randomly tossed n times.
- (i) Suppose $0 \leq r \leq n$.
What is the probability that exactly r 'sixes' appear in the uppermost position? 2
- (ii) **By using the result of part (b), or otherwise**, show that the probability that an 2
odd number of 'sixes' appears is $\frac{1}{2} \left\{ 1 - \left(\frac{2}{3} \right)^n \right\}$
- (d) Let A be the point $(-3, 7)$ and let B be the point $(1, 6)$. Find the coordinates of the 2
point P which divides the interval AB externally in the ratio $1 : 2$.
- (e) A spherical balloon is being deflated so that the radius decreases at a constant rate of 3
8 mm/s. Calculate the rate at which the volume is changing at the instant when the
radius is 100 mm.

- (a) Four women and three men are available for selection in a team. A team of 4 players consists of two men and two women.

(i) How many different teams of four players can be selected? 1

(ii) Two players are husband and wife and wish to play in the same team. How many different teams can now be selected with the husband and wife on the same team? 1

(b)

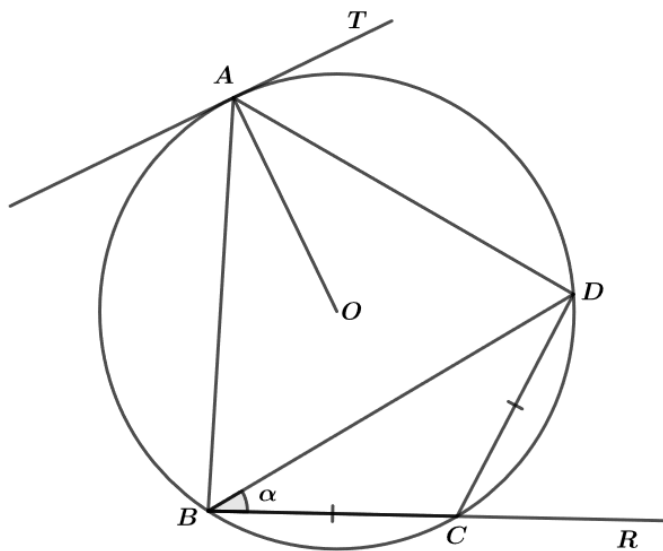


Diagram not to scale

In the diagram above, the points A, B, C and D lie on a circle with centre O . The line TA is a tangent to the circle. The chord BC is produced to R . The interval AO bisects $\angle BAD$ and $BC = CD$. Let $\angle DBC = \alpha$.

(i) Prove that $\angle DCR = 2\alpha$. 1

(ii) Show that $\angle OAD = \alpha$. 1

(iii) Prove that $\angle ABC$ is a right angle. 3

Question 14 is continued on the next page

Question 14 continued

- (c) Let α, β and γ be the roots of $3x^3 + 8x^2 - 1 = 0$. **3**

What is the value of $\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\gamma}\right)\left(\gamma + \frac{1}{\alpha}\right)$?

- (d) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

- (i) Show that the co-ordinates of R , the point of intersection of the normals at P and Q are $(-apq(p+q), a(p^2 + pq + q^2 + 2))$. **2**

- (ii) If $pq = -2$ find the cartesian equation of the locus of R . **3**

End of Examination

Outcomes Addressed in this Question

- PE2 Uses multi-step deductive reasoning in a variety of contexts
PE6 Makes comprehensive use of mathematical language, diagrams & notation for communicating in a wide variety of situations
HE1 Appreciates interrelationships between ideas drawn from different areas of mathematics
HE7 evaluates mathematical solutions to problems and communicates them in an appropriate form

Outcome	Solutions	Marking Guidelines
PE2	(a) Let $t = \tan \frac{\theta}{2}$ Then LHS = $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}$ $= \frac{1 + \frac{2t}{1+t^2} - \left(\frac{1-t^2}{1+t^2}\right)}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$ $= \frac{1+t^2+2t-(1-t^2)}{1+t^2+2t+1-t^2} \text{ (multiplying by } t^2 \text{)}$ $= \frac{2t^2+2t}{2+2t}$ $= \frac{2t(t+1)}{2(1+t)}$ $= t$ $= \tan \frac{\theta}{2}$	2 marks : correct solution 1 mark: significant progress toward correct solution
PE2	Note: a number of students did not state the expression on the LHS, instead starting from the second line. This can result in loss of marks in the HSC. (b) (i) $\cos x - \sqrt{3} \sin x \equiv r \cos(x + \alpha)$ $\equiv r(\cos x \cos \alpha - \sin x \sin \alpha)$ where $r = \sqrt{1^2 + (-\sqrt{3})^2} = 2$ $\cos x - \sqrt{3} \sin x \equiv 2 \cos x \cos \alpha - 2 \sin x \sin \alpha$ Equating like coefficients, $1 = 2 \cos \alpha$, $-\sqrt{3} = 2 \sin \alpha$ $\therefore \sin \alpha = \frac{\sqrt{3}}{2}$, $\cos \alpha = \frac{1}{2}$ As $\sin$ positive quadrants 1, 2 & $\cos$ positive in quadrants 1, 4 α is in quadrant 1. $\therefore \alpha = \frac{\pi}{3}$ $\therefore \cos x - \sqrt{3} \sin x \equiv 2 \cos\left(x + \frac{\pi}{3}\right)$	2 marks : correct solution 1 mark : correct value for r or α
PE2	(ii) Maximum value of $\cos x - \sqrt{3} \sin x$ is when $2 \cos\left(x + \frac{\pi}{3}\right)$ is a maximum which occurs when $\cos\left(x + \frac{\pi}{3}\right) = 1$. $\therefore$ maximum value is $2 \times 1 = 2$.	2 marks : correct answers 1 mark : one correct answer or equivalent

PE2, PE6

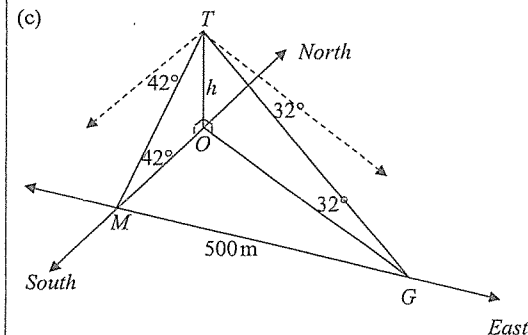
$$\cos\left(x + \frac{\pi}{3}\right) = 1 \text{ when } x + \frac{\pi}{3} = \dots, 0, 2\pi, \dots$$

$$\text{i.e. when } x = \dots, -\frac{\pi}{3}, 2\pi - \frac{\pi}{3}, \dots$$

$\therefore$ maximum value is 2, and the smallest positive

$$\text{value of } x \text{ for which it occurs is } \frac{5\pi}{3}.$$

Note: many students differentiated to find the maximum value. This is not always necessary with trig functions as $\sin$ and $\cos$ have maximum 1.



Let T be the top of the tower.

In Δ 's OTG & OTM , $\cot 32^\circ = \frac{OG}{h}$ & $\cot 42^\circ = \frac{OM}{h}$ respectively

$$\therefore OG = h \cot 32^\circ \text{ \& } OM = h \cot 42^\circ$$

As G is east of M , $\angle OMG = 90^\circ$

$$\therefore OM^2 + 500^2 = OG^2 \text{ (Pythagoras)}$$

$$h^2 \cot^2 42^\circ + 500^2 = h^2 \cot^2 32^\circ$$

$$500^2 = h^2 (\cot^2 32^\circ - \cot^2 42^\circ)$$

$$\therefore h^2 = \frac{500^2}{\cot^2 32^\circ - \cot^2 42^\circ}$$

$$\therefore h = \frac{500}{\sqrt{\tan^2 58^\circ - \tan^2 48^\circ}} \text{ using complementary angle}$$

$$\therefore h = 434 \text{ m results}$$

Note: Many students did not correctly position the angles of depression, or have $\angle OMG$ as the right angle.

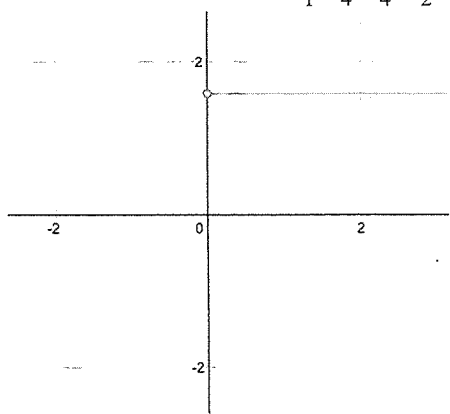
(d) Using $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$,

$$\tan \theta = \left| \frac{\frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right)}{1 + \frac{1}{\sqrt{2}} \times \left(-\frac{1}{\sqrt{2}}\right)} \right|$$

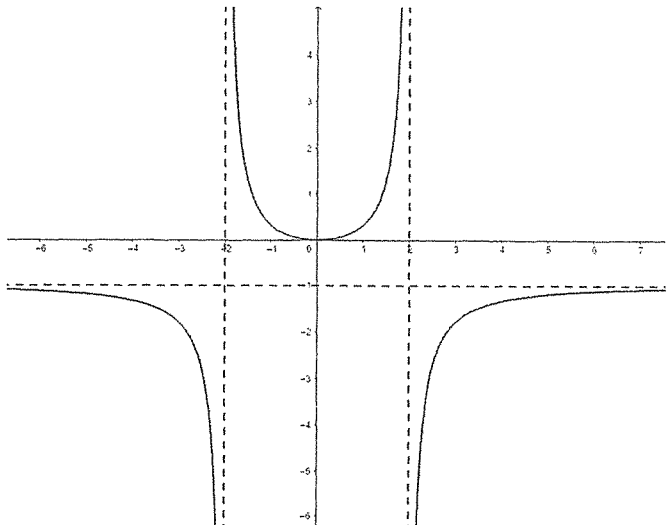
3 marks : correct solution
2 marks: substantial progress towards correct solution
1 mark : significant progress towards correct solution

2 marks : correct solution
1 marks : substantial progress towards correct solution

PE2	$\tan \theta = \left \frac{\sqrt{2}}{\frac{1}{2}} \right $ $\tan \theta = 2\sqrt{2}$ $\theta = 1.23^\circ$	
HE7, HE1	(e) (i) $\tan(x+h) - \tan x = \frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x}$ $= \frac{\sin(x+h)\cos x - \sin x \cos(x+h)}{\cos(x+h)\cos x}$ $= \frac{\sin((x+h)-x)}{\cos(x+h)\cos x}$ $= \frac{\sin h}{\cos(x+h)\cos x}$ (ii) Given $f(x) = \tan x$ and $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, $f'(x) = \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan(x)}{h},$ $= \lim_{h \rightarrow 0} \frac{\sin h}{\cos(x+h)\cos x \cdot h},$ $= \lim_{h \rightarrow 0} \frac{\sin h}{\cos(x+h)\cos x} \times \frac{1}{h},$ $\therefore f'(x) = \lim_{h \rightarrow 0} \frac{\sin h}{h} \times \frac{1}{\cos(x+h)\cos x},$ $= \lim_{h \rightarrow 0} \frac{\sin h}{h} \times \lim_{h \rightarrow 0} \frac{1}{\cos(x+h)\cos x},$ $= 1 \times \frac{1}{\cos(x+0)\cos x},$ $= \frac{1}{\cos^2 x}$ $\therefore f'(x) = \sec^2 x$ In show that questions, do not leave steps out. To ensure full marks, the substitution should be shown and all steps justified, such as $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$.	2 marks : correct solution 1 marks : substantial progress towards correct solution 2 marks : correct solution 1 mark : substantial progress towards correct solution

Year 12	Mathematics Extension 1	Task 4	Examination 2019
Question No.12	Solutions and Marking Guidelines		
Outcomes Addressed in this Questions			
H2 - constructs arguments to prove and justify results			
PE3 - solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations			
HE3 - uses a variety of strategies to investigate mathematical models of situations involving binomial probability, projectiles, simple harmonic motion or exponential growth and decay.			
PE5 - determines derivatives which require the application of more than one rule of differentiation			
HE6 - determines integrals by reduction to a standard form through a given substitution.			
H6 - uses the derivative to determine the features of the graph of a function			
HE 7 - Evaluates mathematical solutions to problems and communicates them in an appropriate form.			
Outcomes	Solutions		Marking Guidelines
PE5	a i) $y = \tan^{-1} u$ $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$ $= \frac{1}{1+u^2} \times \left(-\frac{1}{x^2} \right)$ $= -\frac{1}{1+x^2}$ $u = \frac{1}{x}$ $\frac{du}{dx} = -\frac{1}{x^2}$	2 marks Correct solution with appropriate working 1 mark Error made *note: if answer was inferred from part ii then 1 mark was awarded in part ii and zero for part i	
H2	ii) $\frac{dy}{dx} (\tan^{-1} x + \tan^{-1} \frac{1}{x}) = \frac{1}{1+x^2} - \frac{1}{1+x^2} = 0$	1 mark Correct solution	
H6	iii) $y = \tan^{-1} x + \tan^{-1} \frac{1}{x} \text{ is a constant for } x > 0 \text{ since } \frac{dy}{dx} = 0$ When $x=1$ then $\tan^{-1} 1 + \tan^{-1} \frac{1}{1} = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$ 	1 mark Correct solution	

HE6	b) $x = u^2$ $\frac{dx}{du} = 2u \quad x = 1 \rightarrow u = 1$ $dx = 2u du \quad x = 9 \rightarrow u = 3$ $\int_1^3 \frac{2u du}{u^2 + u} = \int_1^3 \frac{2 du}{u + 1}$ $= [2 \ln(1 + u)]_1^3$ $= 2(\ln 4 - \ln 2)$ $= 2 \ln 2$ $= \ln 4$	3 marks Correct solution 2 marks Error made 1 mark Correctly changed integral limits
PE3	c) $P(1) = 2 - 5 - 9 + 18 = 6$ $P(2) = 16 - 20 - 18 + 18 = -4$ A root lies between $1 < x < 2$ since $P(x)$ is continuous and $P(1)$ and $P(2)$ have opposite signs.	2 marks Correct solution with appropriate reasoning 1 mark Error made or reasoning incomplete
HE3 H2	d i) $N = 125 + Ae^{kt}$ $\frac{dN}{dt} = Ake^{kt}$ $= k(125 + Ae^{kt} - 125)$ $= k(N - 125)$	2 marks Correct solution with appropriate reasoning 1 mark Error made or reasoning incomplete / missing steps
HE3	ii) $25650 = 125 + Ae^0$ $A = 25525$ so $N = 125 + 25525e^{kt}$ $31100 = 125 + 25525e^{5k}$ $\frac{1239}{1021} = e^{5k}$ $\ln \frac{1239}{1021} = 5k \ln e$ $k = \frac{\ln \frac{1239}{1021}}{5} = 0.039$	2 marks Correct solution 1 mark Error made

HE7	e) 	2 marks Correct solution, including smooth graph and correct behaviour of curves at asymptotes 1 mark Error made
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Year 12 2019 Mathematics Extension 1 HSC Task 4-Trial Examination		
Question 13 Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
HE2 - uses inductive reasoning in the construction of proofs H2 - constructs arguments to prove and justify results HE3 - uses a variety of strategies to investigate mathematical models of situations involving binomial probability, projectiles, simple harmonic motion or exponential growth and decay. HE 7 - Evaluates mathematical solutions to problems and communicates them in an appropriate form.		
Outcome	Solutions	Marking Guidelines
HE2	a) Step 1: Test the statement for $n = 1$ $9^{n+2} - 4^n = 729 - 4 = 725$ this is divisible by 5. Step 2: Assume the statement is divisible by 5 for $n = k$ Thus $9^{k+2} - 4^k = 5M$ where M is a positive integer. Step 3: Prove the statement is divisible by 5 for $n = k + 1$. $9^{k+3} - 4^{k+1} = 9(9^{k+2}) - 9(4^k) + 9(4^k) - 4(4^k)$ $= 9(9^{k+2} - 4^k) + 4^k(9 - 4)$ $= 9(5M) + 5(4^k) \dots \dots \dots \text{from step 1}$ $= 5(9M + 4^k)$ $= 5N \quad \text{where } N \text{ is a positive integer.}$ Hence if the statement is divisible by 5 for $n = k$, it is also divisible by 5 for $n = k + 1$. It is divisible by 5 for $n = 1$, so it is divisible by 5 for $n = 2$. If it is true for $n = 2$ it is also true for $n = 3$ and so on. It is then true for all positive integers, n .	3 marks: complete correct proof 2 marks: substantial work that could lead to a correct proof 1 mark: correct working for step 1
HE7, H2	b) (i) $(a+b)^n - (a-b)^n = \left[a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \binom{n}{3}a^{n-3}b^3 + \dots + b^n \right]$ $- \left[a^n - \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 - \binom{n}{3}a^{n-3}b^3 + \dots + (-1)^n b^n \right]$ $= 2 \left[\binom{n}{1}a^{n-1}b + \binom{n}{3}a^{n-3}b^3 + \binom{n}{5}a^{n-5}b^5 + \dots \right]$ $\therefore (a+b)^n - (a-b)^n = 2\binom{n}{1}a^{n-1}b + 2\binom{n}{3}a^{n-3}b^3 + 2\binom{n}{5}a^{n-5}b^5 + \dots$	2 marks: complete correct show 1 mark: for substantial work that could lead to a correct show
HE7, H2	(ii) If n is even, $(-1)^n b^n$ is positive, $\therefore$ last term is $2\binom{n}{n-1}ab^{n-1} = 2nab^{n-1}$.	1 mark: correct last term if n is even

HE3	c) i) $\left(\frac{5}{6} + \frac{1}{6}\right)^n$ is the binomial probability, since probability of a six $= \frac{1}{6}$. $\therefore$ Probability of exactly r sixes, $P(r) = \binom{n}{r} \left(\frac{5}{6}\right)^{n-r} \left(\frac{1}{6}\right)^r$	2 marks: complete correct solution 1 mark: substantial work that could lead to a correct solution
HE3	ii) Probability of an odd number of sixes $= P(1) + P(3) + P(5) + \dots$ $= \binom{n}{1} \left(\frac{5}{6}\right)^{n-1} \left(\frac{1}{6}\right) + \binom{n}{3} \left(\frac{5}{6}\right)^{n-3} \left(\frac{1}{6}\right)^3 + \dots$ $= \frac{1}{2} \left[\left(\frac{5}{6} + \frac{1}{6}\right)^n - \left(\frac{5}{6} - \frac{1}{6}\right)^n \right], \quad \text{from b)}$ $= \frac{1}{2} \left[1 - \left(\frac{2}{3}\right)^n \right]$	2 marks: complete correct solution 1 mark: for substantial work that could lead to a correct solution
HE7	d) $A(-3, 7) \quad B(1, 6)$ $-1:2$ P has coordinates $\left(\frac{(2)(-3) + (-1)(1)}{2-1}, \frac{(2)(7) + (-1)(6)}{2-1} \right)$ $= (-7, 8)$	1 mark: for x coordinate 1 mark: for y coordinate
HE7	e) We are given $\frac{dr}{dt} = -8$ and we wish to find $\frac{dV}{dt}$ $V = \frac{4}{3}\pi r^3$ $\frac{dV}{dr} = 4\pi r^2$ $\therefore \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ $= 4\pi r^2 (-8)$ $= -32\pi r^2$ and when $r = 100$ $\frac{dV}{dt} = -32\pi(100)^2$ $= -320\,000\pi \text{ mm}^3/\text{s}$	3 marks: complete correct solution 2 marks: for finding $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} = 4\pi r^2 (-8)$ 1 mark: for finding $\frac{dV}{dr} = 4\pi r^2$

Q1 B

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \quad (\text{rearranging } \cos 2x = 1 - 2\sin^2 x)$$

$$\int 2 \sin^2 x \, dx$$

$$= \int 2 \left(\frac{1}{2} (1 - \cos(2x)) \right)$$

$$= \int (1 - \cos(2x)) \, dx$$

$$= \int 1 \, dx - \int \cos(2x) \, dx$$

$$= x - \frac{1}{2} \sin 2x + c$$

Q2. D

$$\cos \theta = -\frac{1}{2}$$

$$= \dots, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3} \dots$$

$$\text{or } \theta = 2\pi n \pm \cos^{-1}\left(-\frac{1}{2}\right) = 2\pi n \pm \left(\pi - \cos^{-1}\left(\frac{1}{2}\right)\right)$$

$$= 2\pi n \pm \left(\pi - \frac{\pi}{3}\right)$$

$$\therefore 2\pi n \pm \frac{2\pi}{3}, \text{ where } n \text{ is an integer}$$

Q3. B

Q4 D

$$y = x^3 + 2$$

swap the variables (x and y)

$$x = y^3 + 2$$

Solve the equation for y

$$y^3 = x - 2$$

$$y = \sqrt[3]{x-2}$$

Q5. A

Binomial expansion of $(a + y)^4$

$$= a^4 + 4a^3y + 6a^2y^2 + 4ay^3 + y^4$$

When $a = 2x, y = 3$

$$= 16x^4 - 96x^3 + 216x^2 - 216x + 81$$

$$\therefore -96$$

Q6 C

$$P(2F) = 5C2 (0.48)^2 (0.52)^3$$

$$= \text{combinate } 2F3M \times 2 \text{ Female} \times 3 \text{ males}$$

Q7 B

Sub in $x = 1$ to binomial expansion given

$$\therefore \sum_{r=0}^n nCr = 2^n$$

Q8. A

$$P(-2) = -8 + 4a + 7 = 11$$

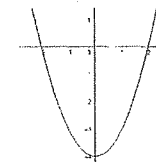
$$4a - 1 = 11$$

$$a = 3$$

Q9. A

$$\frac{x^2-4}{2x} < 0$$

$$(x+2)(x-2) < 0$$



$$0 < x < 2$$

Q10. B

$$5! \times 4! = 2880$$